

ISHIM, Russian Federation

WMO: 285730

Lat: 56.100N

Lon: 69.433E

Elev: 82

StdP: 100.34

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -34.3	(c) -31.1	(d) -36.9	(e) 0.1	(f) -34.2	(g) -33.9	(h) 0.2	(i) -31.0	(j) 8.8	(k) -5.5	(l) 8.1	(m) -6.8	(n) 1.0	(o) 200	(p) 0.622

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.9	(c) 29.7	(d) 19.9	(e) 28.0	(f) 19.1	(g) 26.2	(h) 18.3	(i) 21.2	(j) 27.5	(k) 20.3	(l) 26.0	(m) 19.4	(n) 24.6	(o) 3.5	(p) 150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.1	14.0	23.8	18.2	13.2	23.1	17.2	12.4	22.4	62.1	27.6	58.6	26.0	55.6	24.8	25.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.5	7.5	6.7	-37.2	32.3	3.8	1.6	-39.9	33.5	-42.1	34.4	-44.2	35.3	-46.9	36.5		
			-37.2	22.8	3.7	1.1	-39.9	23.6	-42.1	24.2	-44.2	24.9	-46.9	25.7		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	1.9	-17.5	-15.3	-7.0	3.5	12.0	17.2	18.5	16.2	10.5	3.5	-6.2	-13.8	
	DBStd	14.07	7.76	7.41	6.13	5.75	5.24	4.44	3.65	4.03	4.21	5.02	7.13	8.09	
	HDD10.0	3817	851	709	528	207	40	3	0	2	43	208	486	738	
	HDD18.3	6130	1109	943	787	446	206	72	44	91	239	460	736	997	
	CDD10.0	854	0	0	0	11	101	221	263	195	57	6	0	0	
	CDD18.3	126	0	0	0	0	9	39	49	26	2	0	0	0	
	CDH23.3	1233	0	0	0	4	137	399	436	233	23	0	0	0	
	CDH26.7	303	0	0	0	1	28	114	101	56	3	0	0	0	
Wind	WSAvg	3.1	2.9	3.1	3.5	3.6	3.4	2.8	2.5	2.6	2.8	3.1	3.3	3.3	
Precipitation	PrecAvg	376	15	11	12	22	32	54	68	57	40	28	23	18	
	PrecMax	545	35	33	35	67	116	191	117	114	93	103	47	42	
	PrecMin	224	2	1	0	1	2	13	8	0	8	2	6	0	
	PrecStd	71	8	7	8	16	23	38	29	28	23	20	10	10	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.7	1.9	7.0	22.5	29.6	31.8	31.1	30.5	26.2	19.0	7.3	2.2	
		MCWB	-0.2	0.2	3.1	13.0	16.9	20.3	21.1	20.6	16.7	11.3	5.4	0.9	
	2%	DB	-1.5	-0.2	4.1	18.5	26.7	29.7	29.2	28.2	22.6	16.0	5.2	0.5	
		MCWB	-2.3	-1.2	1.5	10.5	15.7	19.7	20.6	19.5	15.3	9.8	4.0	-0.4	
	5%	DB	-3.6	-2.0	2.6	15.3	24.6	27.7	27.5	26.0	19.8	13.3	3.6	-1.1	
		MCWB	-4.3	-3.1	0.7	9.1	14.7	19.0	19.8	18.6	13.7	8.5	2.3	-2.0	
	10%	DB	-5.8	-4.2	1.3	12.3	21.8	25.5	25.8	23.8	17.6	10.9	2.0	-3.0	
		MCWB	-6.5	-5.1	-0.2	7.4	13.5	17.9	19.1	17.8	12.4	7.6	0.9	-3.7	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.2	0.6	3.6	13.7	18.3	22.0	22.6	21.9	17.8	12.4	5.8	1.1
			MCDB	0.5	1.6	6.5	21.1	26.6	29.6	28.7	28.5	24.2	17.0	6.8	1.7
2%		WB	-2.2	-1.2	1.9	11.0	16.6	20.8	21.4	20.6	15.6	10.7	4.1	-0.4	
		MCDB	-1.5	-0.3	3.6	17.1	24.7	27.7	27.3	26.3	21.4	14.6	5.3	0.5	
5%		WB	-4.3	-3.0	0.9	9.4	15.2	19.7	20.5	19.4	14.4	9.1	2.4	-1.9	
		MCDB	-3.6	-2.2	2.4	15.0	23.3	26.0	26.0	24.4	19.1	12.3	3.4	-1.1	
10%		WB	-6.4	-5.1	-0.1	7.7	13.9	18.6	19.6	18.3	12.9	7.7	1.0	-3.7	
		MCDB	-5.8	-4.2	1.3	12.0	20.5	24.2	24.6	22.9	16.9	10.7	1.8	-3.0	
Mean Daily Temperature Range		5% DB	MDBR	7.8	9.5	9.5	9.4	11.5	10.4	9.9	10.1	9.4	7.4	6.0	7.4
			MCDBR	7.9	8.6	8.4	14.2	15.4	13.6	12.8	13.7	12.9	12.2	5.7	7.1
	MCWBR		7.6	7.9	6.5	8.0	7.0	5.8	5.5	6.4	6.6	7.0	5.0	6.8	
	5% WB	MCDBR	7.7	9.0	7.5	13.3	14.1	12.3	11.4	12.1	12.2	10.8	5.7	6.8	
		MCWBR	7.5	8.4	6.2	7.9	7.1	5.6	5.4	6.1	6.7	6.7	5.1	6.6	
Clear-Sky Solar Irradiance	taub	0.279	0.313	0.369	0.417	0.408	0.408	0.425	0.434	0.381	0.357	0.316	0.282		
	taud	2.043	2.022	1.939	2.064	2.225	2.289	2.239	2.180	2.290	2.213	2.094	1.996		
	Ebn at Noon	587	717	762	791	830	835	810	770	763	667	531	464		
	Edh at Noon	64	98	138	142	128	122	126	126	98	82	61	52		
All-Sky Solar Radiation	RadAvg	0.84	1.87	3.42	4.53	5.40	5.81	5.51	4.30	2.87	1.45	0.69	0.52		
	RadStd	0.06	0.11	0.23	0.39	0.39	0.47	0.47	0.42	0.28	0.23	0.11	0.07		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page